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Gradient Boosted Trees and Operational Regimes for Airport Taxi-Out Time Prediction

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Abstract

Taxi-out time is a common indicator of airport surface efficiency. This paper describes the solution of team *inspiring-kitten* to the EUROCONTROL Performance Review Commission (PRC) Data Challenge 2026, which asked teams to predict departure taxi-out times at ten European airports from 2025 movement and Network Manager (NM) data. The method has three parts. First, an exact identity between the NM and the airport off-block milestones lets gradient boosted trees learn the offset between the two records instead of the raw duration. Second, a deterministic router assigns each departure to one of four operational regimes according to its NM off-block time, which revealed a data issue at Rome Fiumicino, where missing off-block times appear to be filled with the scheduled time. Third, surface-physics features describing the runway queue at pushback, wake separation and taxi routes are added to milestone, traffic and weather features. A controlled out-of-time ablation shows that the offset target improves regular flights by 2.5 s, that the Fiumicino rule alone removes about 105 s of RMSE, and that the physics features add less than 1 s. Departures without a usable NM off-block time are about 1% of traffic but 53% of the squared error. The best public score, 290.14 s RMSE, was reached by blending submissions with an exact quadratic model of the leaderboard score, which is reported here as leaderboard adaptation.

Keywords: Air Traffic Management; taxi-out prediction; airport surface operations; machine learning; gradient boosting

1. Introduction

Departure taxi-out time, the interval between off-block and take-off, is where gate scheduling, apron management, taxiway routing and runway sequencing meet. When departure demand approaches runway capacity, aircraft queue on taxiways with engines running [1]. This increases fuel burn and emissions [2] and makes take-off times harder to predict for the network. Taxi-out estimates are therefore an input to Airport Collaborative Decision Making (A-CDM), where Target Take-Off Times are derived from Target Off-Block Times plus an expected taxi time [3]. They are also used in post-operational performance assessment, where the additional taxi-out time with respect to an unimpeded reference measures surface inefficiency [4].

Predicting taxi-out time is difficult for operational and for informational reasons. On the operational side, the time an aircraft spends between the stand and the runway depends on the departure queue, the runway configuration, the arrival mix on shared runways, the wake-turbulence separation behind the preceding departure [5, 6], the taxi route, and on weather, de-icing and low-visibility procedures. These effects have been modelled with queuing models [1, 2] and, more recently, with regression and machine-learning models fed by surface traffic counts and airport layout features [7, 8, 9]. On the informational side, the timestamps in operational datasets come from different systems with different conventions and resolutions, and these inconsistencies can matter as much as the operational drivers.

The EUROCONTROL Performance Review Commission (PRC) Data Challenge 2026 [10] turned this

problem into an open benchmark. It follows earlier editions of the challenge, among them the 2024 edition on actual take-off weight, for which ensembles of gradient boosted trees and neural networks trained on flight and trajectory data gave accurate open models [11]. Participants received all 2025 movements at ten European airports and had to predict the taxi-out time of the departures in held-out months of 2026. The ranking metric is the root mean squared error (RMSE), which gives a large weight to the rare departures with taxi-out times of several hours.

This paper describes the solution of team *inspiring-kitten*. It is written as a methods paper. The aim is to document what worked and by how much, measured under a controlled protocol, and also what did not work. The contributions are the following.

1. **A milestone-offset formulation of the target.** The challenge target equals the take-off time minus the airport off-block time, while the NM actual off-block time $T^{\mathrm{NM}}_{\mathrm{AOBT}}$ is available at prediction time. Writing $T_{\mathrm{taxi}} = da - \Delta_{\mathrm{offset}}$ with $da = T_{\mathrm{mvt}} - T^{\mathrm{NM}}_{\mathrm{AOBT}}$ lets the models learn the offset between the two off-block records. This target reduces the RMSE on regular departures by 2.5 ± 0.5 s compared with direct regression.
2. **A deterministic operational regime router.** Departures are split into four regimes according to whether $T^{\mathrm{NM}}_{\mathrm{AOBT}}$ is present and valid. This split showed that, at LIRF, departures with no NM off-block time and a large delay against schedule often have a ground-truth taxi-out time equal to the time from the *scheduled* departure to take-off. A one-line rule for these flights, which are 0.03% of departures, removes about 105 s of RMSE on held-out data.
3. **A fast surface-physics feature engine.** For every departure it computes the number of aircraft between pushback and take-off at the moment of its own pushback with an $O(N \log N)$ sweep, a wake-turbulence separation penalty, an unimpeded taxi time from OpenStreetMap routes and the pier contention. A leave-one-group-out ablation of these and of the other feature groups over three seeds shows that only weather has an effect clearly larger than seed noise.
4. **An exact quadratic model of the leaderboard score.** The RMSE of any linear combination of already scored submissions can be predicted from their public scores alone. This model was used to choose blending steps, and the server scores matched its predictions to within about 0.001 s. The paper discusses why this amounts to adaptive use of the public test set [12, 13].
5. **An error decomposition and open release.** In the controlled ablation, the departures without a usable NM off-block time, about 1% of the total, carry 53% of the squared error, which shows where future gains lie. All scripts used to produce the numbers and figures of this paper are released.

Section 2 describes the data, the milestone identity, the features, the regime router and the models. Section 3 presents the controlled ablation, the leaderboard history and the final submission, and Section 4 concludes.

2. Methodology

This section follows the prediction pipeline in the order in which it processes a departure. Section 2.1 presents the data and the evaluation, and Section 2.2 rewrites the target through the milestone identity. Section 2.3 describes the features and Section 2.4 the router that assigns each departure to an operational regime. Section 2.5 gives the learning algorithms and Section 2.6 the score-space blending used on the public leaderboard.

2.1 Problem formulation and data

The challenge data [14] join an airport movement table (suffix `_mvt`) with the NM flight list (suffix `_flt`). The training set contains all 4,167,797 movements of 2025 at Frankfurt (EDDF), Munich (EDDM), London Heathrow (EGLL), Amsterdam Schiphol (EHAM), Barcelona (LEBL), Madrid

(LEMD), Paris Charles de Gaulle (LFPG), Rome Fiumicino (LIRF), Zurich (LSZH) and Istanbul (LTFM), of which 2,085,047 are departures. For a departure i the target is

$$y_i = \texttt{TAXITIME_SEC_mvt} = T_{\mathrm{mvt},i} - T_{\mathrm{block},i}, \tag{1}$$

where T_{mvt} (`MVT_TIME_UTC_mvt`) is the best available take-off time and T_{block} (`BLOCK_TIME_UTC_mvt`) is the off-block time in the airport record. Eq. (1) holds to within one second for every training departure. In the test files T_{block} and y are removed for departures. All other fields remain, including the NM milestones $T^{\mathrm{NM}}_{\mathrm{AOBT}}$ (`AOBT_3_flt`, the actual off-block time of the flown trajectory), `EOBT_1_flt`, `IOBT_flt` and `LOBT_flt`, the scheduled time T_{sched}, the runway, stand, aircraft type, operator and wake category, and the arrivals with their taxi-in times.

The evaluation used two test sets. The *Phase 1* set (`ranking.parquet`) has 344,841 departures in January (152,719) and July (192,122) 2026. It was scored by an automatic evaluation service operated with the OpenSky Network [15], which publishes the RMSE of each submission on a live leaderboard, with a limit of five submissions per team and day [16]. The *Phase 2* set (`final_ranking.parquet`) adds February (144,158) and June (181,791) 2026, for a total of 670,790 departures, and each team submits a single final file for it. The metric is

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}. \tag{2}$$

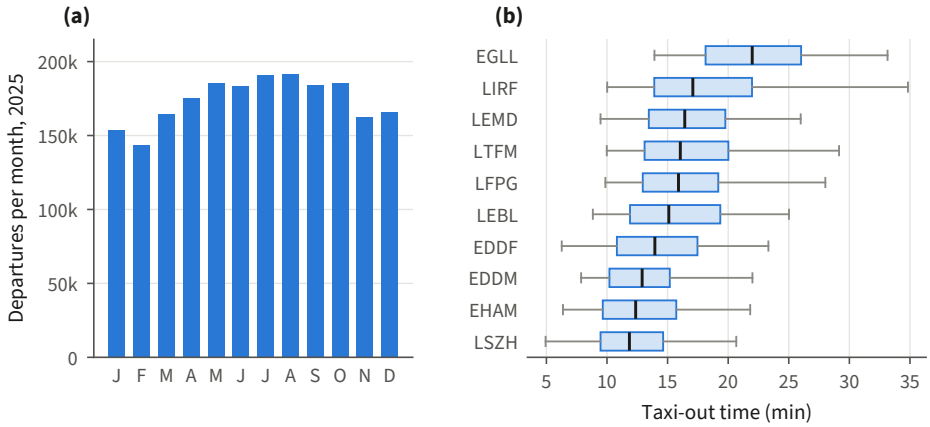


Figure 1. Training data overview. (a) Monthly departures in 2025 across the ten airports. (b) Distribution of taxi-out time per airport (median, inter-quartile range and 5th to 95th percentiles), sorted by median.

Figure 1 shows the seasonal cycle of traffic, from 153.7k departures in January to 191.2k in August, and the spread between airports. The median taxi-out time ranges from about 12 min at LSZH and EHAM to 22 min at EGLL. Over all departures the median is 912 s, the mean 991 s and the standard deviation 546 s. The tail is long, with 584 training departures above two hours and a maximum of 131,167 s (36 h). A model that ignores these flights can still have a good median error and a poor RMSE.

2.2 Operational milestone identity

Because T_{block} is hidden at test time, the most useful available field is the NM off-block time. With

$$da_i = T_{\mathrm{mvt},i} - T^{\mathrm{NM}}_{\mathrm{AOBT},i}, \qquad ds_i = T_{\mathrm{mvt},i} - T_{\mathrm{sched},i}, \tag{3}$$

putting them into Eq. (1) gives

$$y_i = da_i - \Delta_{\text{offset},i}, \qquad \Delta_{\text{offset},i} = T_{\text{block},i} - T^{\text{NM}}_{\text{AOBT},i}. \tag{4}$$

Equation (4) is exact. The offset is the disagreement between the airport and the NM record of the same off-block event. It can come from different sensors and reporting chains, from the difference between the airline's actual off-block message and the airport's detection of pushback, from late updates of the NM time during apron holds, and from rounding. Over the 2025 departures, 97.7% of the $T^{\text{NM}}_{\text{AOBT}}$ values fall exactly on a whole minute, against 5.1% of the take-off times and 6.4% of the airport off-block times, so Δ_{offset} contains a rounding term of up to one minute.

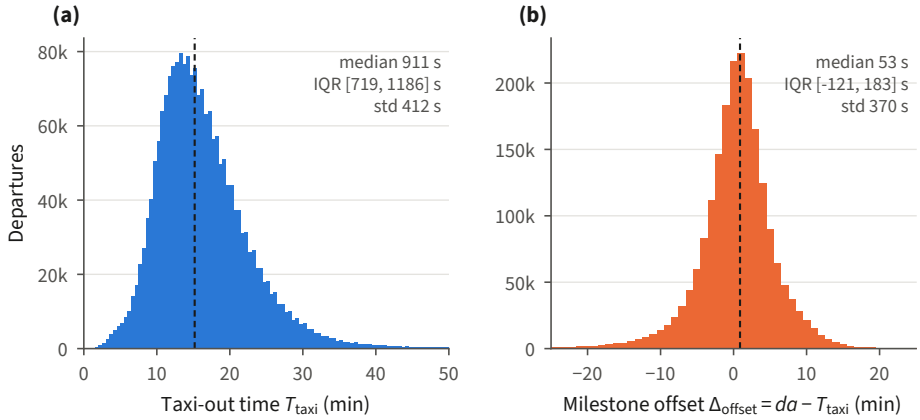


Figure 2. Departures with a valid NM off-block time (Regime 1, 2025). (a) Taxi-out time. (b) Milestone offset $\Delta_{\text{offset}} = da - T_{\text{taxi}}$ in one-minute bins. The offset is close to one minute for most flights and its central half covers about five minutes, but it also has heavy tails.

Figure 2 compares the two quantities for the departures where $T^{\text{NM}}_{\text{AOBT}}$ is present and $120 \le da \le 7200$ s (Regime 1, Section 2.4). The offset has a median of 53 s and an inter-quartile range of $[-121, 183]$ s. Because of these tails, its standard deviation (370 s) is only a little below that of the taxi-out time itself (412 s). The Pearson correlation between da and y in this regime is 0.55 over 2025 and 0.65 in January 2025. The NM off-block time is therefore a reference point and not a substitute for the target. In roughly half of the flights it is within a few minutes of the airport record, and in the rest the difference is large and must be learned. The Regime 1 models are trained on the offset and predict

$$\hat{y}_i = \text{clip}\Big(da_i - \hat{\Delta}_{\text{offset},i},\ 60,\ 7200\Big). \tag{5}$$

With a squared loss, learning Δ_{offset} is not equivalent to learning y with da as a feature when the model is a tree ensemble. Piecewise-constant trees can represent $y = da - c$ only with many splits on da, while the offset target removes this linear part exactly. Section 3.1 measures the difference.

2.3 Feature engineering

All features are computed with Polars and NumPy on each data partition (a single month in the ablation of Section 3.1, a four-month season in production), using only fields present in the test files. Since the evaluation is post-operational, all movements of the partition, including later take-offs and arrival taxi-in times, are visible to the feature extractor. The features in this section therefore describe the traffic situation and are not forecasts available before off-block time. The *base* set has 62 features in six families.

- **Milestones.** *da*, *ds*, a flag for the presence of $T^{\mathrm{NM}}_{\mathrm{AOBT}}$, and the pairwise differences between $T^{\mathrm{NM}}_{\mathrm{AOBT}}$, EOBT, IOBT, LOBT and T_{sched}, clipped to ±2 h.
- **Rolling traffic flow.** Departures per runway and per airport and arrivals per airport in the 15 and 30 min before take-off, all movements per runway in the 15 min before take-off, the mean arrival taxi-in time over 15, 30 and 60 min and its trend, and the time since the previous departure and since the previous movement of either phase on the same runway.
- **Historical encodings.** From the training months, the median, smoothed mean and 15th percentile of taxi time per (airport, stand, runway), and the median per (airport, runway), per (airport, stand) and per (airport, operator).
- **Categorical identifiers and calendar.** Airport, destination, runway, stand and pier, aircraft type, operator, wake category, RECAT-EU category, market segment, flight type, hour (also as sine and cosine), minute and weekday, plus four layout flags such as the use of the EHAM Polderbaan runway.
- **Weather.** Hourly temperature, snowfall, wind speed and a de-icing risk flag from the Open-Meteo historical archive [17], the worst hourly visibility and freezing-fog indicators decoded from METAR reports of the Iowa Environmental Mesonet archive [18], the head-wind and cross-wind components $v\cos(\theta_w - \theta_r)$ and $|v\sin(\theta_w - \theta_r)|$ relative to the departure runway heading θ_r, taken from a table of runway headings in the code, and a tail-wind flag.
- **Taxi route geometry.** The OpenStreetMap distances d_{OSM} and d_{E} and the tortuosity τ described below.

The *physics feature engine* adds nine features, for 71 in total, that target the mechanisms emphasised by queuing models of the departure process [1, 2].

Queue at pushback. For departure i at airport a, the number of other departures of the same airport (and, separately, of the same runway) that are between off-block and take-off at the moment i leaves the stand is

$$Q_i = \sum_{j \neq i,\ \mathrm{ADEP}_j = a} \mathbb{I}\left(T_{\mathrm{push},j} \le T_{\mathrm{push},i} < T_{\mathrm{mvt},j}\right), \tag{6}$$

with $T_{\mathrm{push}} = T^{\mathrm{NM}}_{\mathrm{AOBT}}$, or $T_{\mathrm{mvt}} - 900$ s when $T^{\mathrm{NM}}_{\mathrm{AOBT}}$ is missing. A direct evaluation is $O(N^2)$ per airport. The engine instead builds the $2N$ events (+1 at $T_{\mathrm{push},j}$, −1 at $T_{\mathrm{mvt},j}$), sorts them once, takes the cumulative sum and queries it at every $T_{\mathrm{push},i}$ with a binary search, which gives $O(N \log N)$. The complete physics block, including the features below, takes 0.9 s for one month (307,257 movements) on a laptop CPU.

Wake-turbulence separation. Each aircraft type is mapped to one of the six RECAT-EU categories, from A (super heavy) to F (light) [6], and unmapped types default to E. For consecutive departures on the same runway, a time-based separation $S(c_{\mathrm{lead}}, c_{\mathrm{follow}})$ is taken from Table 1. This is a simplified matrix inspired by the departure separations of RECAT-EU and ICAO Doc 4444 [5], and not the official minima. The features are the penalty $\max(0,\ S - \Delta t)$, where Δt is the observed interval between the two take-offs, the interval Δt itself, and a flag for departures that follow a heavy aircraft (categories A to C).

OpenStreetMap taxi routes. Taxiway, taxilane and runway ways were downloaded from OpenStreetMap [19] and converted into a weighted graph. For every observed (stand, runway) pair the extractor computes the shortest-path distance d_{OSM} from the stand to the runway threshold, the

Table 1. Time-based departure separation matrix $S(c_{\text{lead}}, c_{\text{follow}})$ in seconds used for the wake penalty feature (RECAT-EU categories A to F).

Leader \ Follower	A	B	C	D	E	F
A (super heavy)	100	120	140	160	180	200
B (upper heavy)	90	100	120	140	120	180
C (lower heavy)	90	90	100	120	120	180
D (upper medium)	75	75	75	80	90	120
E (lower medium)	60	60	60	60	70	120
F (light)	60	60	60	60	60	60

straight-line distance d_{E} and the tortuosity $\tau = d_{\text{OSM}}/d_{\text{E}}$. The engine adds an unimpeded taxi time $UTT_{\text{OSM}} = d_{\text{OSM}}/\bar{v}$ with $\bar{v} = 9\ \text{m s}^{-1}$ (about 17.5 kt). Routes were obtained for 3,844 stand-runway pairs at the ten airports, which cover 35% of the 2025 departures. For 10 pairs the graph had no path, and the distance was set to 1.35 times the straight-line distance. Over these departures the correlation between d_{OSM} and taxi-out time is only $r = 0.06$. Taxi distance has been found to be a strong predictor in other studies [7], while Idris et al. [1] found the runway queue to be the main driver. Here the stand and runway identifiers, which are already in the base set, probably carry most of the route information.

Pier contention. Stands whose identifier starts with a letter are grouped by that letter, used as a proxy for the pier or apron area, and the feature counts the pushbacks from the same group in the preceding 15 min. Stands with numeric identifiers are not grouped and receive zero.

Regime indicators. Two binary flags mark the LIRF schedule-filled cases and the extreme apron holds defined in Section 2.4.

2.4 Deterministic operational regime router

The data mix different data-generating processes, and a single model trained on all departures has to find a middle ground between them. Each departure is therefore routed using only da_i, ds_i and the airport, as shown in Figure 3.

- **Regime 1, clean NM off-block time.** $T^{\text{NM}}_{\text{AOBT}}$ is present and $120 \le da_i \le 7200$ s. The prediction comes from the offset ensemble, Eq. (5).
- **Regime 2A, missing or invalid NM off-block time.** $T^{\text{NM}}_{\text{AOBT}}$ is absent, or $da_i < 120$ s, and the departure is not in Regime 2B. In 2025, 4,917 departures have a recorded NM off-block time less than two minutes before take-off, or even after it. A separate LightGBM model is trained on the departures without an NM off-block time (excluding Regime 2B) of four representative months (January, February, June and July 2025), with target $y \in [60, 7200]$ s and without the $T^{\text{NM}}_{\text{AOBT}}$-derived features. It is applied to all Regime 2A departures and its output is clipped to $[60, 3600]$ s.
- **Regime 2B, LIRF schedule fill.** ADEP is LIRF, $T^{\text{NM}}_{\text{AOBT}}$ is absent and $ds_i > 7200$ s. The prediction is $\hat{y}_i = ds_i$, as if the airport off-block time had been set to the scheduled time. In 2025 this rule is exact to within 2 min for 63.3% of the 627 such departures, and 413 of them have a ground-truth taxi-out time above two hours, up to 131,167 s. A likely explanation is that the airport record uses the scheduled time when no actual off-block time was captured. The rule reproduces the target but not the physical taxi duration.

- **Regime 3, extreme hold with an NM off-block time.** $da_i > 7200$ s. Gradient boosted trees cannot extrapolate beyond the training range, so the prediction uses the identity with a fixed offset, $\hat{y}_i = da_i - \tilde{\Delta}_{\mathrm{ADEP}}$. Here $\tilde{\Delta}_{\mathrm{ADEP}}$ is the median offset of the airport among the Regime 1 training rows of the summer models (54 s if unavailable), and the prediction is not clipped at two hours.

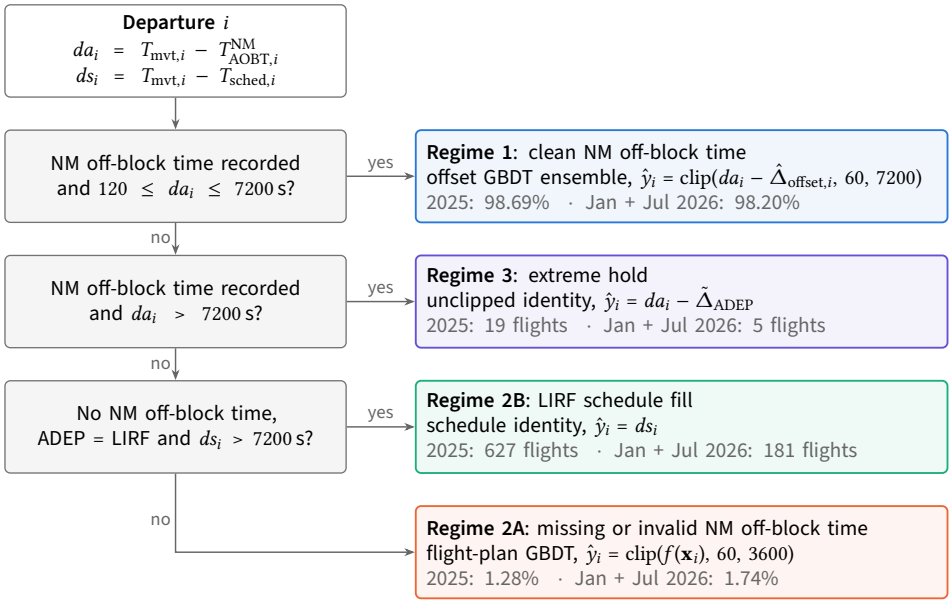


Figure 3. Deterministic regime router. Conditions are tested from top to bottom, and the first that holds assigns the regime. Shares refer to the 2025 training departures and to the Phase 1 test departures (January and July 2026). All predictions are finally rounded to whole seconds and bounded to $[60, 88500]$ s.

Regime 1 covers 98.69% of the 2025 departures and 98.20% of the Phase 1 test departures. Regime 2A covers 1.28% and 1.74%, Regime 2B 627 and 181 flights, and Regime 3 19 and 5 flights. In the Phase 2 test set the corresponding figures are 98.39%, 1.56%, 269 and 8. The Phase 1 test months therefore contain a higher share of missing NM off-block times than the training year, and almost twice the training rate of LIRF schedule-filled departures (0.052% against 0.030%).

The final prediction is rounded to whole seconds and bounded to $[60, 88500]$ s. The upper bound was set after leaderboard probes showed that the test set contains real values of that size (Section 3.3). Since the 2025 maximum is 131,167 s, the bound may still truncate a few extreme cases.

2.5 Learning algorithms

Regime 1 is predicted by a weighted average of three gradient boosted decision tree (GBDT) implementations trained on the offset, a choice in line with the open take-off weight models of the 2024 edition of the challenge [11].

- LightGBM [20] with leaf-wise growth, 127 leaves, depth 9, learning rate 0.035, 1,000 trees, at least 40 samples per leaf and column subsampling of 0.80.
- XGBoost [21] with the histogram method on an NVIDIA RTX 4060 Laptop GPU, depth 9, learning rate 0.035, 900 trees, row and column subsampling of 0.85 and 0.80, L_1 and L_2 regularisation of 1 and 3, and native categorical handling.
- CatBoost [22] with oblivious trees of depth 8 on GPU, learning rate 0.045, 900 iterations and or-

dered target statistics for the categorical identifiers.

The offsets are averaged with weights 0.40, 0.35 and 0.25. Separate winter and summer models are trained on January, February, November and December 2025 and on May to August 2025, and are applied to the matching test months. Training rows are restricted to Regime 1 with $60 \le y \le 7200$ s and $|\Delta_{\text{offset}}| \le 3600$ s. During Phase 1, tabular residual networks with entity embeddings for the categorical fields [23] and multilayer perceptrons were also trained. Individually they were weaker than the GBDTs, as is common on tabular data [24], but they added diversity to the blends described next.

2.6 Score-space quadratic blending

Let $\hat{\mathbf{y}}_a$ and $\hat{\mathbf{y}}_b$ be two submissions with public scores S_a and S_b on the same N test departures, and let $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ denote the error vectors. Expanding $\|\hat{\mathbf{y}}_a - \hat{\mathbf{y}}_b\|^2 = \|\mathbf{e}_a - \mathbf{e}_b\|^2$ gives

$$\frac{1}{N}\,\mathbf{e}_a^{\top}\mathbf{e}_b = \frac{1}{2}\left(S_a^2 + S_b^2 - \frac{1}{N}\|\hat{\mathbf{y}}_a - \hat{\mathbf{y}}_b\|^2\right). \tag{7}$$

The right-hand side contains only public scores and the submitted vectors, so the full error Gram matrix $\mathbf{G}$ of a set of scored submissions can be computed without knowing $\mathbf{y}$. For any weights $\mathbf{w}$ with $\sum_j w_j = 1$, the blend $\sum_j w_j \hat{\mathbf{y}}_j$ has error $\sum_j w_j \mathbf{e}_j$, and its squared score is exactly $\mathbf{w}^{\top}\mathbf{G}\mathbf{w}/N$. Blending then becomes a small quadratic program, similar to stacking [25, 26], except that the holdout is the public test set itself. Along a single direction $\boldsymbol{\delta} = \hat{\mathbf{y}}_b - \hat{\mathbf{y}}_a$ the squared score is a parabola,

$$\mathcal{L}^2(k) = \mathrm{RMSE}^2(\hat{\mathbf{y}}_a + k\boldsymbol{\delta}) = S_a^2 + 2kc + k^2 q, \qquad q = \frac{\|\boldsymbol{\delta}\|^2}{N}, \quad c = \frac{S_b^2 - S_a^2 - q}{2}, \tag{8}$$

with its lowest point at $k^* = -c/q$ and minimum $\mathcal{L}^2(k^*) = S_a^2 - c^2/q$. Two scored submissions determine the whole curve. Rounding to integer seconds and clipping make the identity approximate. For blends made only of already scored submissions, the predicted and the returned scores never differed by more than about 0.001 s.

Equation (7) is the regression version of known attacks on leaderboards that report aggregate scores [13]. Every step chosen in this way uses information from the public test labels. The gains it brings therefore measure adaptation to the Phase 1 set and should not be read as generalisation [12]. The method was used on purpose, because the challenge ranks teams on their best public score, and its contribution is reported separately from that of the models.

3. Results and discussion

The results are presented in two parts. Sections 3.1 and 3.2 measure the contribution of each component under a controlled protocol on held-out 2025 data, without leaderboard feedback. Sections 3.3 and 3.4 report the public leaderboard history and the final submission, and the section closes with negative results and threats to validity.

3.1 Controlled ablation on held-out data

Leaderboard scores mix model quality with blending choices and cannot show what each component adds. A controlled out-of-time experiment was therefore run with the script `ablation_dec_jan.py` of the repository. All models are trained on December 2025 and evaluated on all 153,706 departures of January 2025, the same month as the winter half of the Phase 1 test set but one year earlier. Each configuration uses one LightGBM model (63 leaves, learning rate 0.05, 450 trees, row and column subsampling of 0.8), so that the effect of each component can be isolated. The Regime 2A model

of row (d) is a LightGBM with 300 trees trained on the December Regime 2A departures, with the base features and the target clipped to $[60, 3600]$ s. The Regime 3 offset is the median December offset, 55 s. Each configuration is trained with three random seeds, and the tables report the mean ± standard deviation. In January 2025 the regimes contain 152,064 (Regime 1), 1,617 (2A), 24 (2B) and 1 (3) departures. The absolute errors cannot be compared with the production results, which use four training months, three libraries and other test months, but the differences between rows are meaningful.

Table 2. Cumulative ablation on January 2025, with models trained on December 2025. RMSE in seconds, mean ± s.d. over three seeds. "All" covers the 153,706 departures and "Regime 1" the 152,064 departures with a valid NM off-block time. In rows (b) and (c) the departures outside Regime 1 receive a constant 1,050 s, and row (a) predicts all departures with its own model.

Configuration	All	Regime 1
(a) Direct GBDT on y, base features, no router	453.38 ± 0.59	234.44 ± 0.55
(b) Offset GBDT on Δ_{offset}, base features	475.12 ± 0.10	231.93 ± 0.20
(c) (b) + physics feature engine	474.74 ± 0.46	231.14 ± 0.95
(d) (c) + Regime 2A model	471.99 ± 0.35	231.14 ± 0.95
(e) (d) + Regime 2B LIRF schedule identity	366.93 ± 0.45	231.14 ± 0.95
(f) (e) + Regime 3 unclipped identity (full router)	367.24 ± 0.45	231.14 ± 0.95
Removing one router component from (f)		
without the Regime 2B rule	472.23 ± 0.35	
without the Regime 2A model (constant 1,050 s)	370.77 ± 0.59	
without the Regime 3 rule (constant 1,050 s)	366.93 ± 0.45	

Table 2 leads to four observations.

The offset target helps on regular flights. Comparing (a) and (b) on Regime 1, learning Δ_{offset} instead of y lowers the RMSE by 2.51 ± 0.46 s, consistently across seeds. On all departures (b) is worse than (a) only because (b) assigns a constant to the departures outside Regime 1, which shows how much these few departures weigh in the metric.

The LIRF rule dominates the router. Adding the Regime 2B identity for 24 departures (0.016% of the month) lowers the all-flight RMSE by about 105 s, from 471.99 to 366.93 s. The Regime 2A model contributes 2.8 s. The Regime 3 rule makes the result slightly worse here (+0.31 s). January 2025 contains a single such departure, with da = 7523 s and a taxi-out time of 1,493 s, so its NM off-block time is about 100 min earlier than the airport record. The identity misses it by 5,975 s, against 443 s for the constant. The rule was kept because leaderboard probes showed that capping predictions for extreme cases is costly on the test set (Section 3.3), but this holdout does not show its benefit.

Physics features help little, and the effect is within noise. Going from (b) to (c) lowers the Regime 1 RMSE by 0.80 ± 1.15 s. Table 3 removes one feature group at a time from configuration (f). Only weather has an effect clearly above seed variability, with +2.43 s on Regime 1. Queue at pushback (+0.71 s), wake separation (+0.61 s) and pier contention (+0.48 s) have the expected sign but are within one or two standard deviations. The OpenStreetMap distances have an effect close to zero, and removing the historical encodings or the rolling-flow counts does not hurt, which suggests that their information is already covered by the stand, runway and milestone features. A likely reason for the small effect of the queue features is that da already includes the queue the aircraft actually went through, since it spans the same interval. Queue features would matter more in a predictive setting where T_{mvt} is unknown.

Table 3. Leave-one-group-out ablation from configuration (f) of Table 2. Δ is the increase in RMSE (s) when the group is removed, so positive values mean that the group helps. Mean ± s.d. over three seeds.

Removed group	Features	Δ RMSE, Regime 1	Δ RMSE, all
Weather (Open-Meteo, METAR, runway wind)	9	$+2.43 \pm 0.86$	$+1.52 \pm 0.53$
Queue at pushback, Eq. (6)	2	$+0.71 \pm 0.83$	$+0.44 \pm 0.52$
Wake separation (Table 1, RECAT class)	4	$+0.61 \pm 0.65$	$+0.38 \pm 0.40$
Pier contention	1	$+0.48 \pm 1.24$	$+0.30 \pm 0.77$
OpenStreetMap taxi network	4	$+0.10 \pm 0.92$	$+0.06 \pm 0.57$
Rolling traffic flow (15, 30 and 60 min)	14	-0.29 ± 1.14	-0.18 ± 0.71
Historical encodings	6	-1.02 ± 0.99	-0.64 ± 0.62

The remaining error sits where the NM off-block time is missing. Under configuration (f), the Regime 1 departures (98.9% of the month) account for 39.2% of the squared error, the Regime 2A departures (1.05%) for 53.3%, the 24 Regime 2B departures for 7.3% and the single Regime 3 departure for 0.2%. The RMSE within Regime 2A is about 2,615 s. In 2025, 704 of the 26,760 Regime 2A departures have $y > 3600$ s and 69 exceed two hours. The flight-plan features available for these departures barely distinguish them from normal ones, and the clip at 3,600 s removes large errors on the majority while leaving the tail unexplained.

3.2 Feature importance

Figure 4 shows the share of total split gain for configuration (c) with seed 42. The model relies mostly on the NM milestones, `aobt_minus_eobt` with 25.9%, *da* with 21.7% and `aobt_minus_iobt` with 5.3%, on the stand (9.1%), and on the destination, runway, operator and stand-runway encodings. Since the target is the difference between two off-block records, it makes sense that milestone differences pointing to late NM updates or changes of the ready time are useful. All physics-engine features rank 29th or lower, the highest being the runway queue at pushback with 0.23% of the gain. Split gain is a biased importance measure and does not equal predictive contribution, but it agrees with the ablation of Table 3.

3.3 Public leaderboard history

Table 4 lists selected submissions to the Phase 1 leaderboard. Some were diagnostic probes rather than attempts to improve the score, and these were among the most useful.

Three lessons follow from this history. The largest single improvement (v2 to v7, −123 s) came from separating departures by the state of their NM off-block time, which was the first version of the router of Section 2.4. The probes v10 (capping at 18,000 s, +97 s with respect to v9) and v49 (softening the extreme predictions, +6 s) showed that the Phase 1 labels contain real taxi-out values of many hours and that a few departures can move the score by tens of seconds, so the tail predictions were kept unclipped up to 88,500 s. Finally, from v25 onwards almost all improvements came from blending, and the last 9 s between the best non-blended pipeline (v91, 299.39 s) and the best score (v112, 290.14 s) were obtained by score-space blending.

Figure 5 shows the last step. With $\hat{\mathbf{y}}_a$ = v104 ($S_a = 290.7445$), $\hat{\mathbf{y}}_b$ = v110 ($S_b = 294.6465$) and $q = 1039.71$ s^2 computed from the two files, Eq. (8) gives $c = 622.24$ s^2, a vertex at $k^* = -0.598$ and a predicted minimum of 290.1034 s. A shorter step, $k = -0.45$, was submitted as a safety margin against the effects of rounding, which the parabola does not model. For this step the curve predicts 290.1429 s, and the server returned 290.1429 s. The optimal step is negative, which means moving

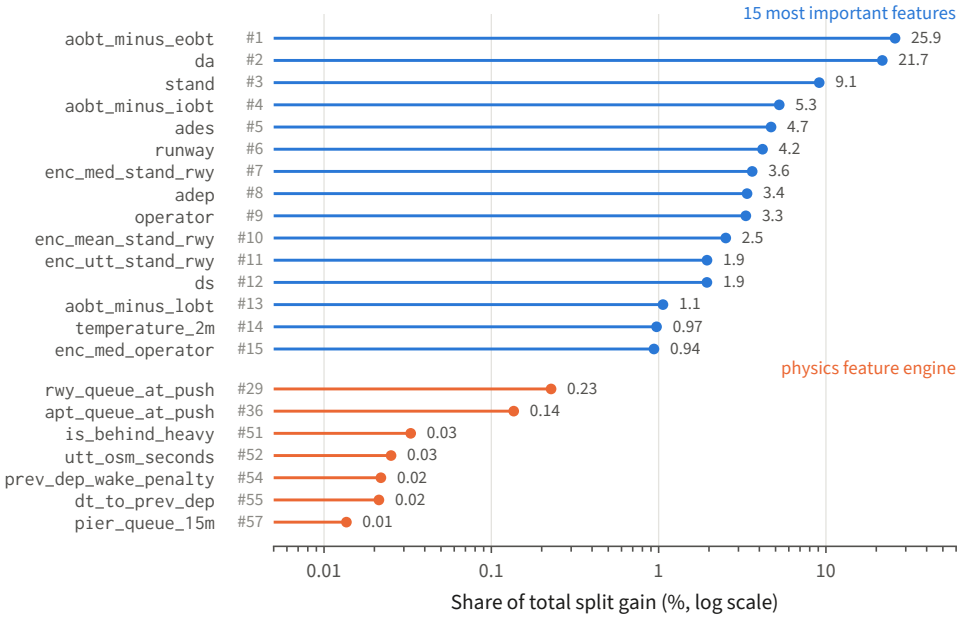


Figure 4. Share of total split gain for the LightGBM offset model of configuration (c) (71 features, December 2025 training, seed 42). The 15 most important features are shown in blue at the top and every physics-engine feature in orange at the bottom. Grey labels give the overall rank, and the horizontal axis is logarithmic.

away from the physics-engine model. This reflects that the model (v109, 326.04 s alone) was weaker on the public months than the blend it was mixed with.

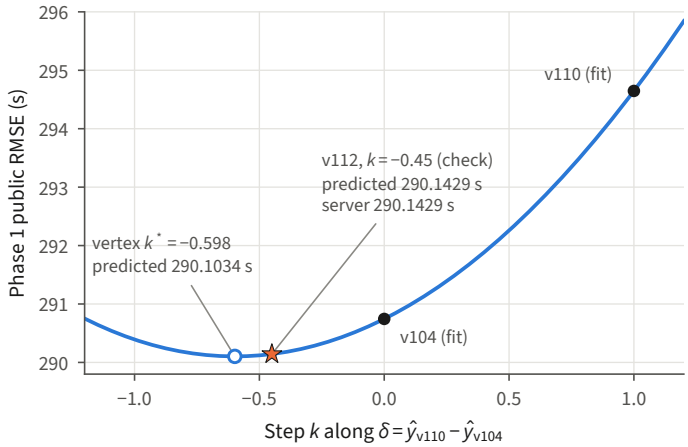


Figure 5. Score-space parabola of Eq. (8) along $\boldsymbol{\delta} = \hat{\mathbf{y}}_{\mathrm{v110}} - \hat{\mathbf{y}}_{\mathrm{v104}}$. The curve is fixed by the two public scores at $k = 0$ and $k = 1$ and by $\overline{\delta^2}$. Submission v112 at $k = -0.45$ was not used for the fit and lies on the curve.

Given the error decomposition of Section 3.1, further gains are most likely to come from the departures without a usable NM off-block time and from the extreme tail, and much less from the regular departures, where every feature group changed the RMSE by less than 3 s.

Table 4. Selected Phase 1 submissions (344,841 departures, January and July 2026). The score is the public RMSE in seconds.

Version	Date	Content	Score
v1	15 Sep	LightGBM residual on $da-60$ s, predictions clipped to $[120, 5400]$	545.50
v2	15 Sep	Direct LightGBM, target clipped to $[60, 15000]$	476.33
v7	16 Sep	First regime split, with separate models with and without NM off-block time	353.45
v9	16 Sep	v8 retrained with weather and runway sequencing features	345.50
v10	16 Sep	Probe with predictions capped at 18,000 s	442.75
v14	17 Sep	External data (OurAirports, Open-Meteo, RECAT-EU)	314.84
v21	18 Sep	Convex blend with CatBoost, extreme predictions kept from v18	308.64
v25	20 Sep	GBDT stack and tabular neural network	299.26
v47	22 Sep	Blend weight at the vertex of Eq. (8) (predicted 294.529)	294.53
v49	23 Sep	Probe with extreme predictions softened (50% blend)	300.65
v60	24 Sep	METAR and aircraft features, blend of two feature generations	293.42
v91	30 Sep	Best non-blended pipeline, seasonal LightGBM and CatBoost	299.39
v104	2 Oct	Quadratic program over six scored submissions, Eq. (7)	290.74
v109	12 Oct	Physics-engine GBDT ensemble alone	326.04
v110	12 Oct	Blend of v104 and v109 (50/50 on Regime 1)	294.65
v111	12 Oct	Production pipeline of Section 2.5 alone	319.79
v112	12 Oct	Step $k = -0.45$ along v110 − v104 (predicted 290.1429)	290.14

3.4 Phase 2 final submission

The final file (`inspiring-kitten_final.parquet`, 670,790 departures) combines two sources. For January and July 2026 (344,841 departures) it contains the v112 predictions. For February and June 2026 (325,949 departures), where no public score is available, it contains the production pipeline of Section 2.5 with the regime router. The two halves were therefore produced differently. The January and July half benefits from leaderboard-guided blending, whose gains are specific to those months. The February and June half comes from a pipeline whose stand-alone public score on January and July was 319.79 s. If February and June behave like January and July, the final RMSE should lie between these two values. The Phase 2 result was not available at the time of writing.

Before upload, the file passed an automated consistency check. It has exactly 670,790 rows in the order of the official template, no missing values, integer type, all predictions within $[60, 88500]$ s, a mean of 1,045.5 s with a standard deviation of 534.6 s, and only three predictions at the lower bound. This check protects against formatting and clipping errors, such as a bug in an intermediate version (v109) that left 915 predictions at the 60 s lower bound, but it says nothing about accuracy.

3.5 Negative results

Several ideas that seemed promising did not help, and they are reported here so that they need not be repeated. Per-airport specialist models generalised worse than global models on a December holdout (319.98 s against 307.58 s) and on the leaderboard (v37, 308.93 s). A stacking meta-learner over the specialists was worse still, with 344.30 s on the same holdout. Separate blend weights for winter and summer were slightly worse than a single weight (v50, +0.17 s). The Open-Meteo archive used does not provide visibility, and its fog codes were always absent for these airports, so the visibility and

low-visibility features came only from METAR. Low cloud cover from the reanalysis had almost no correlation with taxi-out time (r = 0.009). OpenStreetMap route distance, despite its physical appeal, carries almost no signal once the stand and runway are known (Table 3).

3.6 Threats to validity

The controlled experiment uses one training month and a single model family, so its absolute numbers differ from those of the production ensemble. The Regime 2B and Regime 3 effects rest on 24 and 1 held-out departures, so their size is uncertain, although the 2025 statistics support the LIRF rule. Several hyper-parameters, clipping bounds and blend weights were chosen after observing public scores, so the public scores in Table 4 are too optimistic as estimates of the performance on new months [12].

4. Conclusion

This paper described a taxi-out prediction pipeline for the PRC Data Challenge 2026 built on an exact milestone identity, a deterministic regime router, a set of surface-physics features and gradient boosted tree ensembles. A controlled out-of-time ablation separated what mattered from what did not. Learning the offset between the airport and NM off-block records instead of the raw duration consistently improved regular departures. A single rule for schedule-filled departures at LIRF had the largest effect on RMSE of any component. Weather features helped, while the queue, wake, pier and OpenStreetMap features contributed little once the NM milestones were available. The remaining error is concentrated in the 1% of departures without a usable NM off-block time and in the extreme tail. The best public score was 290.14 s, and its last improvements came from an exact score-space blending procedure, documented here as leaderboard adaptation and not as model improvement.

Two lessons follow for A-CDM and performance monitoring. The first is that the quality and consistency of off-block milestones, in their coverage, their resolution and the convention used when no actual time is captured, affect taxi-out statistics as much as operational drivers do. A fallback to the scheduled time, as observed at LIRF, can turn a missing timestamp into a taxi-out record of several hours. The second is that queue and separation features are most useful when predicting *before* take-off. In the post-operational setting of the challenge their information is largely contained in the observed take-off time. Future work should model the missing-milestone regime directly, for example by recovering the off-block time from ADS-B surface tracks of the OpenSky Network [15]. The same tracks could measure the actual taxi path and the time spent in the runway queue, and the features should also be evaluated in a real-time setting where T_{mvt} is unknown.

Author contributions

- Joao P. A. Dantas. Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Software, Validation, Visualization, Writing (Original Draft), Writing (Review and Editing).

Open data and reproducibility statement

The challenge data are provided by the EUROCONTROL Performance Review Commission to registered participants [14]. Weather, airport, aircraft-type and taxiway data come from open sources [18, 17, 27, 28, 19], and the processed files are included in the repository together with the scripts that download them. The code is released under the GNU General Public License v3 (GPLv3) at https://github.com/jpadantas/PRC-Challenge-2026. The repository contains the feature extraction (`src/features_fast.py`, `src/features_physics_engine.py`), the production pipeline of Sections 2.4 and 2.5 (`src/train_and_predict_final_phase.py`), and the construction of v110, v112

and the final file from the versioned v104 and v109 (`src/generate_v110.py`, `src/generate_v112.py`, `src/update_final_submission_with_v112.py`). It also contains the ablation script and its outputs for Tables 2 and 3 (`joas_paper/scripts/ablation_dec_jan.py`) and the script that regenerates the figures and the dataset statistics from the raw data (`joas_paper/scripts/make_figures.py`). The scripts behind the earlier submissions, including v104, are kept with the full submission log in `archive/`.

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